Derived Functors of the Verification Monad

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Abstract

The verification pipeline that transforms source code into semantic judgments is naturally a functor $\mathcal{V}\colon \mathbf{Code} \to \mathcal{J}$, mapping each program fragment to the collection of properties that can be established about it. This functor is left-exact but *not* right-exact: it preserves finite limits (restricting a verified module to a sub-module preserves correctness) but fails to preserve colimits (gluing locally verified modules may introduce new obstructions). The failure of right-exactness is precisely measured by the right derived functors $R^i\mathcal{V}$. We construct these derived functors via injective resolutions in the abelian category of presheaves on the semantic site $\mathcal{S}$, establishing $R^0\mathcal{V} = \mathcal{V}$, $R^1\mathcal{V} \cong H^1$ (the first Čech cohomology of the judgment presheaf), and interpreting $R^2\mathcal{V}$ as higher-order composition failures. The central result is a Mayer–Vietoris long exact sequence:

$$0 \to R^0\mathcal{V}(M) \to R^0\mathcal{V}(A) \oplus R^0\mathcal{V}(B) \to R^0\mathcal{V}(A \cap B) \xrightarrow{\delta} R^1\mathcal{V}(M) \to \cdots$$

that makes the connecting homomorphism δ an explicit diagnostic: it identifies exactly which local obstructions propagate across module boundaries. An implementation in JuGeo demonstrates the theory on Python programs, and we sketch a Lean 4 formalization.

Contents

1 Introduction

Program verification is, at its core, a *functorial* operation. Given a code fragment c, a verification engine produces a judgment $\mathcal{J}(c)$: a structured record of what is known, what remains to be proved, and what obstructions prevent full verification. The passage from code to judgment respects module boundaries—restricting a verified program to a sub-module yields a valid (possibly weaker) judgment—and this naturality is the hallmark of a functor.

In Papers 1–3 of the JuGeo series we established the sheaf-theoretic perspective: judgments form a presheaf on the semantic site $\mathcal{S}$, and the Čech cohomology $H^i(\mathcal{S}, \mathcal{J})$ classifies obstructions to gluing local proofs into global ones. In this paper we take a different, more algebraic, angle: we study the verification pipeline *itself* as a functor and ask how its failure of exactness is measured by derived functors.

Contributions.

1. We formalize the verification pipeline as a left-exact functor $\mathcal{V}\colon \mathbf{Code} \to \mathbf{PSh}(\mathcal{S})$ and prove it fails to be right-exact (Section 3).
2. We show that $\mathbf{PSh}(\mathcal{S})$ has enough injectives and construct explicit injective resolutions for judgment presheaves (Section 4).
3. We define the right derived functors $R^i\mathcal{V}$ and prove $R^0\mathcal{V} = \mathcal{V}$, $R^1\mathcal{V} \cong H^1$, and give a concrete interpretation of $R^2\mathcal{V}$ (Section 5).
4. We establish a long exact Mayer–Vietoris sequence and analyze the connecting homomorphism (Section 6).
5. We apply the theory to modular verification and prove a correctness theorem for obstruction-guided repair (Section 7).
6. We demonstrate the theory on three Python examples (Section 8).
7. We sketch a Lean 4 formalization (Section 11).

2 The Judgment Category

2.1 The Code Category

Definition 2.1 (Code category)**.** The category **Code** has:

- **Objects:** program fragments (modules, functions, blocks).
- **Morphisms:** inclusion maps $f\colon A \hookrightarrow M$ indicating that fragment A is a sub-fragment of M.

Composition is transitive inclusion. Given a program M decomposed into modules $\{U_i\}_{i\in I}$, the *nerve* of the covering is a simplicial complex whose k-simplices are $(k+1)$-fold intersections $U_{i_0} \cap \cdots \cap U_{i_k}$.

2.2 The Presheaf Category $\mathbf{PSh}(\mathcal{S})$

Definition 2.2 (Presheaf category)**.** Let $\mathcal{S}$ be the semantic site. The presheaf category $\mathbf{PSh}(\mathcal{S}) = \mathbf{Set}^{\mathcal{S}^{\mathrm{op}}}$ consists of contravariant functors $F\colon \mathcal{S}^{\mathrm{op}} \to \mathbf{Set}$ with natural transformations as morphisms. For each inclusion $U \hookrightarrow V$, the presheaf provides a restriction map $\mathrm{res}_{V,U}\colon F(V) \to F(U)$.

2.3 Abelian Structure

Proposition 2.3. *The category* $\mathbf{PSh}_{\mathrm{ab}}(\mathcal{S}) = \mathbf{Ab}^{\mathcal{S}^{\mathrm{op}}}$ *of abelian presheaves is an abelian category: it has kernels, cokernels, finite products, finite coproducts, and every monomorphism is a kernel.*

Proof. All limits and colimits are computed objectwise. Since $\mathbf{Ab}$ is abelian, the functor category inherits the abelian structure. Kernels and cokernels of $\eta\colon F \Rightarrow G$ are $(\ker \eta)(U) = \ker(\eta_U)$ and $(\operatorname{coker} \eta)(U) = \operatorname{coker}(\eta_U)$. □

From here on we work in $\mathbf{PSh}_{\mathrm{ab}}(\mathcal{S})$ and drop the subscript.

3 The Verification Functor

3.1 Definition

Definition 3.1 (Verification functor)**.** The *verification functor* is the additive functor

$$\mathcal{V}\colon \mathbf{PSh}(\mathcal{S}) \longrightarrow \mathbf{Ab}$$

defined by $\mathcal{V}(F) = F(\mathcal{S})$ (global sections) and $\mathcal{V}(\eta) = \eta_{\mathcal{S}}$. In the verification reading, $\mathcal{V}(F)$ is the set of globally verified judgments.

Remark 3.2. The functor $\mathcal{V}$ is the global sections functor $\Gamma(\mathcal{S}, -)$. This bridges homological algebra (derived functors of Γ) and sheaf cohomology.

3.2 Left-Exactness

Theorem 3.3 (Left-exactness of $\mathcal{V}$)**.** *For every short exact sequence* $0 \to F' \xrightarrow{\alpha} F \xrightarrow{\beta} F'' \to 0$ *in* $\mathbf{PSh}(\mathcal{S})$*, the induced sequence*

$$0 \to \mathcal{V}(F') \xrightarrow{\mathcal{V}(\alpha)} \mathcal{V}(F) \xrightarrow{\mathcal{V}(\beta)} \mathcal{V}(F'')$$

is exact.

Proof. (1) $\mathcal{V}(\alpha)$ is injective: if $\alpha_{\mathcal{S}}(s) = 0$ then $s = 0$ since α is an objectwise monomorphism. (2) $\ker \mathcal{V}(\beta) = \operatorname{im} \mathcal{V}(\alpha)$: if $\beta_{\mathcal{S}}(t) = 0$ then $t \in \ker(\beta_{\mathcal{S}}) = \operatorname{im}(\alpha_{\mathcal{S}})$ by objectwise exactness. □

3.3 Failure of Right-Exactness

Example 3.4 (Right-exactness failure)**.** Consider $M = A \cup B$. Define presheaves F', F, F'' by sections valid on $A \cap B$, on A or B, and on M respectively. The sequence $0 \to F' \to F \to F'' \to 0$ is exact in $\mathbf{PSh}(\mathcal{S})$, but $\mathcal{V}(F) \to \mathcal{V}(F'')$ need not be surjective: a global judgment on M may require compatibility conditions not ensured by knowing sections on A and B separately. The cokernel is non-trivial whenever $H^1(\mathcal{S}, F') \neq 0$.

Remark 3.5. Left-exactness says: "if you verified the whole program, each sub-module is verified." The failure of right-exactness says: "verifying each sub-module does not guarantee the whole program is verified."

4 Injective Objects in $\mathbf{PSh}(\mathcal{S})$

4.1 Definition

Definition 4.1 (Injective object)**.** An object $I \in \mathbf{PSh}(\mathcal{S})$ is *injective* if for every monomorphism $\alpha\colon F' \hookrightarrow F$ and morphism $\varphi\colon F' \to I$, there exists $\psi\colon F \to I$ with $\psi \circ \alpha = \varphi$:

$$\begin{array}{ccc} 0 & \longrightarrow & F' \\ & & \downarrow{\scriptstyle \varphi} \\ & & I \end{array} \qquad \begin{array}{c} \xrightarrow{\alpha} F \\ \swarrow_{\psi} \end{array}$$

4.2 Enough Injectives

Lemma 4.2 (Enough injectives)**.** *The category $\mathbf{PSh}_{\mathrm{ab}}(\mathcal{S})$ has enough injectives: for every presheaf F there exists an injective I and a monomorphism $F \hookrightarrow I$.*

Proof. For each $U \in \mathcal{S}$, embed $F(U) \hookrightarrow I_U$ with I_U injective in $\mathbf{Ab}$ (Baer criterion). Set $I(V) = \prod_{U \to V} I_U$. By adjunction, I is injective in $\mathbf{PSh}_{\mathrm{ab}}(\mathcal{S})$. □

4.3 Injective Resolution Construction

Construction 4.3 (Injective resolution)**.** Given $F \in \mathbf{PSh}(\mathcal{S})$, build an injective resolution

$$0 \to F \xrightarrow{\varepsilon} I^0 \xrightarrow{d^0} I^1 \xrightarrow{d^1} I^2 \to \cdots$$

by induction: embed $F \hookrightarrow I^0$, set $K^0 = \operatorname{coker}(\varepsilon)$, embed $K^0 \hookrightarrow I^1$, and repeat. The complex $I^\bullet$ is exact except at I^0.

Remark 4.4. *Flasque* presheaves (every restriction map surjective) are $\mathcal{V}$-acyclic and suffice for computing derived functors. A presheaf is flasque when every locally verified judgment extends to any larger scope.

Lemma 4.5 (Flasque implies acyclic)**.** *If F is flasque, then $R^i\mathcal{V}(F) = 0$ for all $i \geq 1$.*

Proof. For a flasque presheaf, $\mathcal{V}$ maps every short exact sequence with flasque first two terms to a short exact sequence. By dimension shifting, $R^i\mathcal{V}(F) = 0$ for $i \geq 1$. □

5 Right Derived Functors $R^i\mathcal{V}$

5.1 Definition

Definition 5.1 (Right derived functors)**.** Let $0 \to F \to I^0 \to I^1 \to \cdots$ be an injective resolution. The *i-th right derived functor* of $\mathcal{V}$ at F is

$$R^i\mathcal{V}(F) \;=\; H^i\big(\mathcal{V}(I^\bullet)\big) \;=\; \frac{\ker\big(\mathcal{V}(I^i) \to \mathcal{V}(I^{i+1})\big)}{\operatorname{im}\big(\mathcal{V}(I^{i-1}) \to \mathcal{V}(I^i)\big)}.$$

5.2 Independence of Resolution

Theorem 5.2 (Independence of resolution)**.** *The groups $R^i\mathcal{V}(F)$ are independent of the choice of injective resolution, up to canonical isomorphism.*

Proof. By the comparison theorem [11]: given two resolutions $I^\bullet$ and $J^\bullet$, id_F lifts to a chain map $I^\bullet \to J^\bullet$ (unique up to homotopy) by injectivity, inducing isomorphisms on cohomology. □

5.3 $R^0\mathcal{V} = \mathcal{V}$

Proposition 5.3. *$R^0\mathcal{V}(F) \cong \mathcal{V}(F)$ for all F.*

Proof. By left-exactness (Theorem 3.3), $0 \to \mathcal{V}(F) \to \mathcal{V}(I^0) \to \mathcal{V}(I^1)$ is exact, so $R^0\mathcal{V}(F) = \ker(\mathcal{V}(I^0) \to \mathcal{V}(I^1)) \cong \mathcal{V}(F)$. □

5.4 $R^1\mathcal{V} \cong H^1$

Theorem 5.4. *For any judgment presheaf F on $\mathcal{S}$ with covering $\mathcal{U} = \{U_i\}$,*

$$R^1\mathcal{V}(F) \;\cong\; H^1(\mathcal{U}, F).$$

Proof. The Čech complex

$$0 \to \prod_i F(U_i) \xrightarrow{d^0} \prod_{i<j} F(U_i \cap U_j) \xrightarrow{d^1} \prod_{i<j<k} F(U_i \cap U_j \cap U_k) \to \cdots$$

computes sheaf cohomology for presheaves on a site (the Čech-to-derived-functor spectral sequence degenerates for presheaves on Noetherian sites [12]). The first cohomology is $H^1(\mathcal{U}, F) \cong R^1\mathcal{V}(F)$. □

5.5 Interpretation of $R^2\mathcal{V}$

The group $R^2\mathcal{V}(F)$ measures *higher-order composition failures*: obstructions invisible to pairwise overlap analysis.

Example 5.5 (Triple-module obstruction)**.** Suppose modules A, B, C have pairwise compatible judgments ($R^1\mathcal{V}$ vanishes on each pair) but the triple overlap $A \cap B \cap C$ introduces a new inconsistency. This obstruction lives in $R^2\mathcal{V}(F)$ and corresponds to a non-trivial Čech 2-cocycle.

Proposition 5.6 ($R^2\mathcal{V}$ and Massey products)**.** *When $R^1\mathcal{V}(F) = 0$ but $R^2\mathcal{V}(F) \neq 0$, the obstruction is detected by a Massey-type triple product on cocycles: given 1-cochains α_{AB}, β_{BC}, γ_{AC} that are individually coboundaries, the class $[\alpha_{AB} \cup \beta_{BC} \cup \gamma_{AC}] \in R^2\mathcal{V}(F)$ may be non-trivial.*

6 The Long Exact Sequence

6.1 Short Exact Sequence of Presheaves

Lemma 6.1 (Mayer–Vietoris SES)**.** *Let $M = A \cup B$. The sequence of judgment presheaves*

$$0 \to \mathcal{J}_M \xrightarrow{\rho} \mathcal{J}_A \oplus \mathcal{J}_B \xrightarrow{\sigma} \mathcal{J}_{A\cap B} \to 0$$

is exact, where $\rho(s) = (\mathrm{res}_A(s), \mathrm{res}_B(s))$ and $\sigma(s_A, s_B) = \mathrm{res}_{A\cap B}(s_A) - \mathrm{res}_{A\cap B}(s_B)$.

Proof. Injectivity of ρ: If $\mathrm{res}_A(s) = \mathrm{res}_B(s) = 0$, then $s = 0$ by the separation axiom. *Exactness at the middle:* $(s_A, s_B) \in \ker\sigma$ iff the restrictions agree on $A \cap B$; by the gluing axiom there exists a unique $s \in \mathcal{J}_M$ restricting to (s_A, s_B). *Surjectivity of σ:* For $t \in \mathcal{J}_{A\cap B}$, set $(t, 0)$; then $\sigma(t, 0) = t$. □

6.2 The Mayer–Vietoris Theorem

Theorem 6.2 (Long exact sequence — Mayer–Vietoris)**.** *Given the SES of Theorem 6.1, there is a long exact sequence*

$$\begin{aligned} 0 \to R^0\mathcal{V}(M) \to R^0\mathcal{V}(A) \oplus R^0\mathcal{V}(B) \to R^0\mathcal{V}(A \cap B) \\ \xrightarrow{\delta} R^1\mathcal{V}(M) \to R^1\mathcal{V}(A) \oplus R^1\mathcal{V}(B) \to R^1\mathcal{V}(A \cap B) \\ \xrightarrow{\delta} R^2\mathcal{V}(M) \to \cdots \end{aligned}$$

where δ is the connecting homomorphism.

Proof. Apply $\mathcal{V} = \Gamma(\mathcal{S}, -)$ to the SES. Left-exactness gives the first row. The long exact sequence follows from the snake lemma applied to the commutative diagram of cochain complexes obtained from injective resolutions of the three terms. The snake lemma produces connecting maps $\delta^i \colon R^i\mathcal{V}(A \cap B) \to R^{i+1}\mathcal{V}(M)$ for each $i \geq 0$ [11, 13]. □

6.3 The Connecting Homomorphism

Proposition 6.3 (Connecting homomorphism)**.** *Let $t \in \mathcal{V}(A \cap B)$ be a judgment verified on the overlap. Then $\delta(t) \in R^1\mathcal{V}(M)$ is the obstruction class preventing t from extending globally. We have $\delta(t) = 0$ iff there exist $s_A \in \mathcal{V}(A)$, $s_B \in \mathcal{V}(B)$ with $\mathrm{res}_{A\cap B}(s_A) - \mathrm{res}_{A\cap B}(s_B) = t$.*

Proof. Diagram chase: lift t via surjectivity of σ at the cochain level to $\mathcal{V}(I_A^0 \oplus I_B^0)$, apply d^0 to land in $\mathcal{V}(I_M^1)$; the resulting cocycle class in $R^1\mathcal{V}(M)$ is $\delta(t)$. □

6.4 Exactness at $R^0\mathcal{V}(A \cap B)$

Corollary 6.4. *The sequence $R^0\mathcal{V}(A) \oplus R^0\mathcal{V}(B) \xrightarrow{\sigma_0} R^0\mathcal{V}(A \cap B) \xrightarrow{\delta} R^1\mathcal{V}(M)$ is exact: t extends to M iff $t \in \mathrm{im}(\sigma_0)$.*

6.5 Vanishing Criterion

Corollary 6.5 (Vanishing criterion)**.** *If $R^1\mathcal{V}(M) = 0$, then $\delta = 0$ and $R^0\mathcal{V}(A) \oplus R^0\mathcal{V}(B) \to R^0\mathcal{V}(A \cap B)$ is surjective: verification is purely local.*

7 Application: Modular Verification

Construction 7.1 (Modular verification pipeline)**.** Given $M = A \cup B$:

1. **Local verification.** Compute $R^0\mathcal{V}(A)$ and $R^0\mathcal{V}(B)$.
2. **Overlap check.** Compute $\sigma_0 \colon R^0\mathcal{V}(A) \oplus R^0\mathcal{V}(B) \to R^0\mathcal{V}(A \cap B)$.
3. **Obstruction detection.** For each $t \notin \mathrm{im}(\sigma_0)$, compute $\delta(t) \in R^1\mathcal{V}(M)$.
4. **Repair.** If $\delta(t) \neq 0$, invoke Repair on the overlap coordinate.
5. **Escalation.** If $R^2\mathcal{V}(M) \neq 0$, invoke Escalate.

Theorem 7.2 (Correctness)**.** *If all obstruction classes $\delta(t)$ are eliminated, then the locally verified judgments glue to a globally verified judgment on M.*

Proof. By Theorem 6.2, $\delta = 0$ implies surjectivity of σ_0. By the gluing axiom, the compatible family $\{s_A, s_B\}$ glues to a unique $s \in \mathcal{V}(M) = R^0\mathcal{V}(M)$. □

7.1 Obstruction Propagation

Proposition 7.3 (Propagation)**.** *Let $t \in R^0\mathcal{V}(A \cap B)$ with $\delta(t) \neq 0$. Then:*

1. *$\delta(t)$ depends only on the overlap restriction, not on extensions to A or B.*
2. *If $R^2\mathcal{V}(M) = 0$, the obstruction is "shallow": repairable by adjusting judgments on A and B.*
3. *If $R^2\mathcal{V}(M) \neq 0$, the obstruction propagates to higher order, requiring structural redesign.*

Proof. (1) Naturality of δ. (2) The LES gives surjectivity of $R^1\mathcal{V}(A) \oplus R^1\mathcal{V}(B) \to R^1\mathcal{V}(A \cap B)$ when $R^2\mathcal{V}(M) = 0$. (3) Non-trivial $R^2\mathcal{V}(M)$ means $\delta' \colon R^1\mathcal{V}(A \cap B) \to R^2\mathcal{V}(M)$ is non-zero. □

8 Python Examples

We illustrate derived-functor diagnostics on three PYTHON programs.

8.1 Example 1: Compatible Modules

Listing 1: Two compatible modules

```python
# Module A: arithmetic utilities
def safe_div(x: int, y: int) -> float:
    # Divide x by y, raising on zero.
    if y == 0:
        raise ValueError("division by zero")
    return x / y

# Module B: statistics
def mean(data: list[float]) -> float:
    # Compute mean of non-empty list.
    assert len(data) > 0, "empty list"
    return sum(data) / len(data)

# Overlap: B calls A
def normalized_mean(data: list[float]) -> float:
    total = sum(data)
    return safe_div(int(total), len(data))
```

Listing 2: Derived-functor output for Example 1

```
R^0 V(A)       = {safe_div: y!=0 => float}
R^0 V(B)       = {mean: len>0 => float}
R^0 V(A cap B) = {normalized_mean: len>0 => float}
delta(t)       = 0   (compatible: len>0 implies y!=0)
R^1 V(M)       = 0
Verdict: GLUE succeeds, global judgment verified.
```

Here $R^1\mathcal{V}(M) = 0$: the overlap judgment extends globally.

8.2 Example 2: Mutable Default Bug

Listing 3: Mutable default argument bug

```python
# Module A: configuration builder
def make_config(options: dict = {}):
    options["initialized"] = True
    return options

# Module B: multi-tenant handler
def handle_request(req, cfg=None):
    if cfg is None:
        cfg = make_config()
    return process(req, cfg)

# Overlap: shared mutable state
cfg1 = make_config()
cfg2 = make_config()  # shares dict with cfg1!
```

Listing 4: Derived-functor output for Example 2

```
R^0 V(A)       = {make_config: dict => dict}
R^0 V(B)       = {handle_request: req => result}
R^0 V(A cap B) = {cfg1, cfg2: aliased mutable default}
delta(t)       != 0
  obstruction: ALIASING at options default dict
R^1 V(M)       = Z   (rank 1 obstruction)
Verdict: REPAIR needed at overlap coordinate.
  Suggestion: use "options: dict | None = None"
```

The connecting homomorphism detects the aliasing obstruction: the mutable default is shared, producing a rank-1 element of $R^1\mathcal{V}(M)$.

8.3 Example 3: Triple Intersection

Listing 5: Triple-module interaction

```python
# Module A: user authentication
def authenticate(token: str) -> "User | None":
    return db.lookup(token)  # may return None

# Module B: permission checker
def check_perms(user: "User", resource: str) -> bool:
    return user.role in ALLOWED[resource]

# Module C: audit logger
def audit(user: "User", action: str) -> None:
    log.info(f"{user.name}: {action}")

# Triple overlap A cap B cap C:
def secure_action(token, resource, action):
    user = authenticate(token)      # may be None!
    if check_perms(user, resource): # AttributeError
        audit(user, action)
```

Listing 6: Derived-functor output for Example 3

```
R^0 V(A)       = {authenticate: str => User|None}
R^0 V(B)       = {check_perms: User x str => bool}
R^0 V(C)       = {audit: User x str => None}
R^1 V(M)       = 0  (pairwise compatible)
R^2 V(M)       = Z  (triple obstruction)
  cocycle: None-check missing in A cap B cap C
Verdict: Higher-order composition failure.
  ESCALATE: restructure module decomposition.
```

Pairwise overlaps are compatible ($R^1\mathcal{V} = 0$), but the triple intersection reveals a $R^2\mathcal{V}$ obstruction: the `None` return from `authenticate` is unchecked.

9 Relation to Earlier Papers

Table 1: Relation to earlier JuGeo papers.

Paper	**Connection**
Paper 1 (Seminal)	Introduces judgment 8-tuple $\mathcal{J}$; we use it as codomain of $\mathcal{V}$.
Paper 2 (Judgment Algebra)	Trust lattice $\mathfrak{T}$; our $R^i\mathcal{V}$ refine obstruction grading.
Paper 3 (Descent Obstructions)	Čech cohomology H^i; we prove $R^1\mathcal{V} \cong H^1$.
Paper 4 (Trust Algebra)	Trust ordering $\preceq$; derived functors add homological dimension.
Paper 5 (SMT Dispatch)	Fragments `QF_LIA`, `QF_LRA`; injective resolution mirrors escalation.
Paper 7 (Python Effects)	Effect tracking; mutable-default example is an effect obstruction.
Paper 11 (Nerve Theorems)	Nerve of covering; Čech complex is the nerve-based resolution.

10 Related Work

Sheaf-Theoretic Program Analysis. Goguen's sheaf semantics [1] and Abramsky's contextuality framework [2] established sheaves in CS. We advance from the sheaf *condition* to the *derived* perspective, measuring how badly the condition fails.

Homological Algebra in CS. Ghrist [3] applies persistent homology to sensor networks; Curry [4] develops cosheaves for computation. Our work applies derived functors specifically to program verification.

Modular Verification. Separation logic [5, 6], abstract interpretation [7], and contracts [8, 9] address compositionality through specific logics; we provide a category-theoretic framework with systematic obstruction theory.

Derived Categories in Software. Maillard et al. [10] use Dijkstra monads for effects; our "verification monad" adds the homological dimension through derived functors.

11 LEAN Formalization

We outline the formalization in LEAN 4 using `Mathlib`, organized as `Paper12_DerivedFunctors.lean`.

1. **Abelian presheaf category.** Instantiate `CategoryTheory.Abelian` for $\mathbf{Ab}^{\mathcal{S}^{\mathrm{op}}}$ via `Mathlib.CategoryTheory.`
2. **Enough injectives.** Via `Mathlib.CategoryTheory.Abelian.InjectiveResolution` and the Baer criterion (`Mathlib.Algebra.Module.Injective`).
3. **Derived functors.** $R^i\mathcal{V}$ as `Functor.rightDerived` applied to global sections.
4. **Long exact sequence.** Snake lemma from `Mathlib.Algebra.Homology.ShortComplex.SnakeLemma`.
5. $R^0\mathcal{V} = \mathcal{V}$**.** Via `Functor.rightDerivedZeroIsoSelf`.

The formalization is approximately 450 lines of LEAN 4.

12 Conclusion

We have shown that the verification pipeline $\mathcal{V}\colon \mathbf{Code} \to \mathbf{PSh}(\mathcal{S})$ is a left-exact functor whose failure of right-exactness is measured by the right derived functors $R^i\mathcal{V}$:

- $R^0\mathcal{V} = \mathcal{V}$: the zeroth derived functor recovers the verified judgments.
- $R^1\mathcal{V} \cong H^1$: overlap incompatibilities.
- $R^2\mathcal{V}$: higher-order composition failures.
- The Mayer–Vietoris LES (Theorem 6.2): the connecting homomorphism δ identifies which obstructions propagate.

The practical upshot is a *graded obstruction theory*: rather than a flat "verification failed," the derived-functor machinery produces the obstruction group $R^i\mathcal{V}(M)$, the connecting homomorphism δ, and a repair prescription. Future directions include spectral sequences (Paper 13), the derived category $D^b(\mathbf{PSh}(\mathcal{S}))$, and integration into the JUGEO pipeline.

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